

Peter Massih

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Education

Master's in Computer Science, Specialization in Data Analytics and Machine Learning, EPFL, Switzerland	2023 - 2026
Exchange year, Computer Science and Electrical Engineering, ETHZ, Switzerland	2022 - 2023
Bachelor's in Computer and Communication Systems Engineering, EPFL, Switzerland	2020 - 2023

Work Experience

Software engineer AI/ML, Internship - Lombard Odier & Cie SA, Geneva, Switzerland	Oct 2025 – Present
- Deployed a production anomaly detection system (ElasticNet regression) to flag statistically significant outliers across large-scale financial tax data. - Built a CPU-only inference pipeline (llama.cpp) fusing macro indicators, equity research signals, and portfolio analytics to generate long/short recommendations. - Re-architected and extended the on-prem LLM RAG orchestration system (LlamaIndex, Elasticsearch), cutting query latency from 8 min to 20s and indexing time by 80%. Integrated into production bank APIs via Java/Spring Boot and Kafka pipelines.	
Machine Learning Researcher, Internship - NEC Laboratories America, Princeton, NJ, USA	Feb 2025 – Oct 2025
- Developed QVLM, a code-generation vision-language architecture achieving +13.9 % accuracy vs. GPT-5. First-author paper on arXiv, under review at ECCV 2026. - Created the SQuID benchmark comprising 2,000 Q&A pairs from NAIP and remote sensing datasets with human-validated answer ranges. - Ablated code generators and trained segmentation backends against GPT-5 and VLM baselines.	
Trader Career Kickstarter - Optiver, Amsterdam, Netherlands	Apr 2025
- Selected for competitive trading program (1% acceptance rate). Built a market-making bot in Python for the Optibook challenge. - Competed in real-time market-making simulations and options pricing theory sessions.	
Full-Stack AI Engineer - Geospatial, Lausanne, Switzerland	Jul 2024 – Oct 2024
- Developed an LLM framework (FastAPI, LangChain, LangGraph), integrating GPT-4, Claude Sonnet, and Llama 3.1 70B to orchestrate function-calling plans for natural-language geospatial queries, deployed on Azure. - Launched an interactive web mapping app, adopted by swisstopo and the Swiss Agency for Development and Cooperation, delivered via Agile sprints in JIRA.	
Student Teaching Assistant - EPFL, Lausanne, Switzerland	
Java Programming (Fall 2023, Fall 2024), Computer Systems in C (Spring 2024)	

Projects and Theses

Using Diffusion Models to generate Brick-Compatible 3D Objects (Diffusers, PyTorch, Trimesh, NumPy)
- Engineered a two-stage conditional diffusion pipeline to generate 3D LEGO models from text prompts, using a transformer-based UNet3D conditioned on CLIP text embeddings. - Voxelized 542k samples from the Objaverse dataset, improving model quality by 57%. - Designed a greedy LegoLization algorithm to convert voxel outputs into buildable brick models, open-sourced code, model, and datasets.
Wikipedia Navigation Analysis (Python, Pandas, Statsmodels, NetworkX)
- Analyzed 75,000+ Wikipedia gameplay paths to evaluate four navigation strategies (hub usage, semantic similarity, link position, backtracking) using PageRank, TF-IDF cosine similarity, and Spearman correlation - Built a Mixed Linear Model with random effects and backward selection to control for game-difficulty confounders, identifying semantic navigation as the dominant strategy, decreasing average completion time by 21 seconds ($p < 0.001$, $n = 50,138$).
Reinforcement Learning Mountain Car (Python, Gymnasium, PyTorch)
- Devised and benchmarked deep Q networks (DQN), random-network-distillation (RND), and model-based Dyna agents. - Introduced a direction-aware throttle reward, accelerating DQN convergence by ~33%, reducing training from 300 to 200 episodes.

Skills

Technical: Python, C, Java, Scala, R | PyTorch, scikit-learn, Spark | Spring Boot, Angular, Kafka | Docker, Kubernetes, Elasticsearch, SQL | Azure | Git | GraphQL

Coursework: Stochastic Calculus, Probability & Statistics, Optimization, Statistical Learning, Advanced Algorithms.

Language: English (Native), French (Native), Arabic (Native), and German (Limited Working Proficiency)

Leadership and Extracurricular Activities

Lebanese Swimming Federation — Silver medal (butterfly) and bronze medal (freestyle) at national competitions	2016 – 2021
Model United Nations Delegate (New York and Lebanon) — Awarded Research & Position Paper prize	2018 – 2020
Red Cross Youth Section (Lebanon) — Led awareness campaigns on first aid and humanitarian topics among Lebanese youth	2018 – 2020